Measures of location and spread

This is identical to the equivalent chapter in the old S1 module. Some content will be familiar from GCSE (mean of grouped/ungrouped data), but many concepts new (e.g. standard deviation) along with possibly unfamiliar notation.

1: Mean, Median, Mode

"Calculate the mean of this grouped frequency table."

2: Quartiles, Percentiles, Deciles

"Use linear interpolation to estimate the interquartile range."

3: Variance & Standard

"Calculate the standard deviation of the maths marks."

4: Coding

"The marks x where coded using $y = 2x + 10$. Given that the standard deviation of y is 5, determine the standard deviation of x."

Variables in algebra vs stats

$$x$$

Similarities

- Just like in algebra, variables in stats **represent the value of some quantity**, e.g. shoe size, height, colour.
- As we saw in the previous chapter, variables can be discrete or continuous.
- **Can be part of further calculations**, e.g. if x represents height, then $2x$ represents twice people's height. In stats this is known as '**coding**', which we'll cover later.

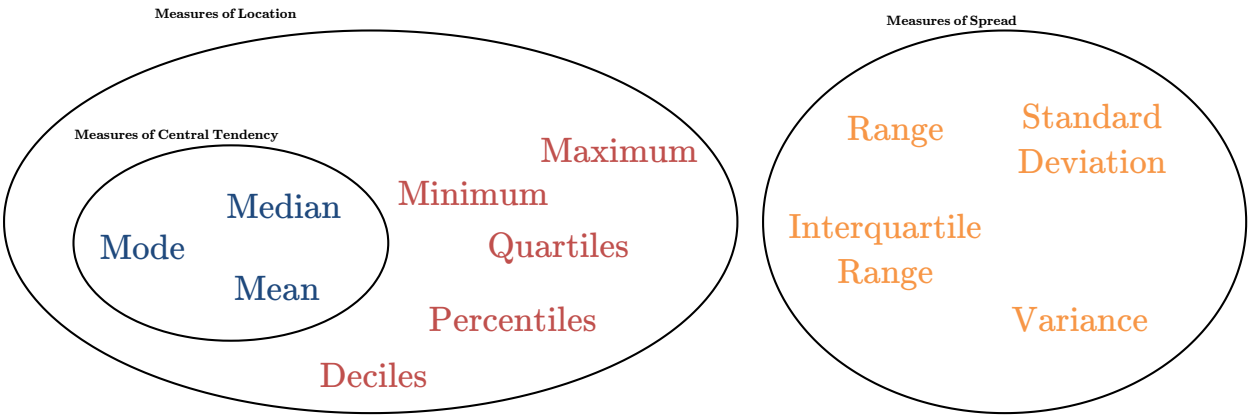
Differences

- Unlike algebra, a variable in stats represents the value of **multiple objects** (i.e. it's a bit like a set). e.g. the heights of **all** people in a room.
- Because of this, we can do **operations** on it as if it was a **collection of values**:
  - If x represents people's heights,

    $$\Sigma x$$

    gives the sum of everyone's heights. In algebra this would be meaningless: if $x = 4$, then Σx makes no sense!
  - $\overline{x}$ is the mean of x. Notice x is a collection of values whereas $\overline{x}$ is a single value.
- To each value of the variable, **we could attach an associated probability**. This is known as a **random variable** (Chapter 6).

Measure of..



Measures of location are single values which describe a **position** in a data set.

Of these, **measures of central tendency** are to do with the **centre of the data**, i.e. a notion of 'average'.

Measures of spread are to do with **how data is spread out**.

x	162	178	150	160	160	170

You all know how to find the mean of a list of values. But let's consider the notation and see how theoretically we could calculate each of the individual components on a calculator.

$$\bar{x} = \frac{1}{2}$$

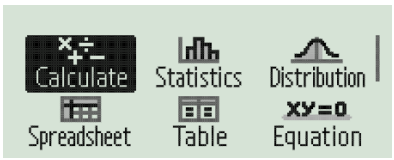
The 'overbar' in stats specifically means the sample mean of x

x	162	178	150	160	160	170

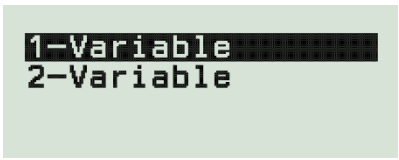
These are instructions for the **Casio fx-570/991CW**



1. Use the arrows and OK to select Statistics.



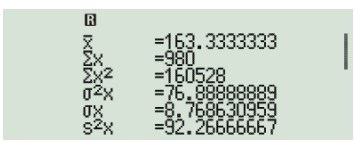
2. Choose 1-Variable.



3. Enter each value with EXE. After the last entry press OK to open the menu.



4. Select **1-Var Results** and read off the values of Σx and n and $\overline{x}$.

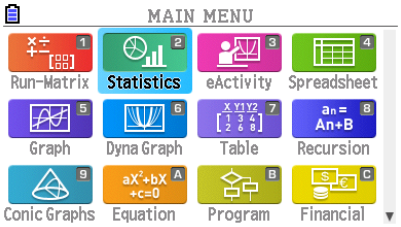


x	162	178	150	160	160	170

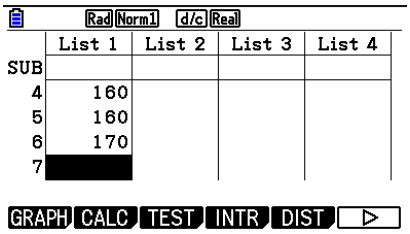
These are instructions for the **Casio fx-CG50**



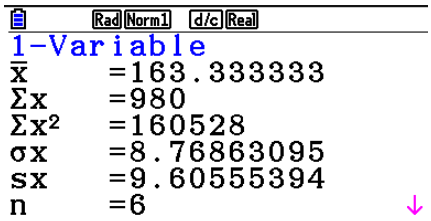
1. Choose 2 for Statistics.



2. Enter the values, pressing EXE after each.



3. Press CALC (F2) then 1-VAR (F1). Read off the values $\overline{x}$ and Σx.

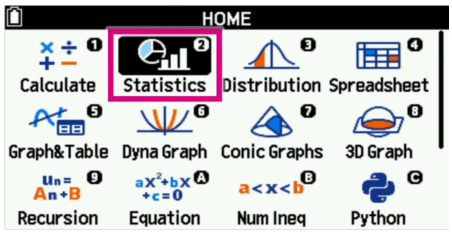


x	162	178	150	160	160	170

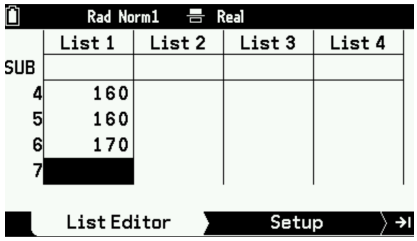
These are instructions for the **Casio fx-CG100**



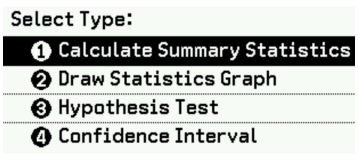
1. Choose 2 for Statistics.



2. Enter the values, pressing EXE after each.

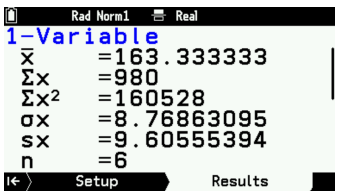


3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and 1 for Freq. Select Execute.





4. Read off the values $\overline{x}$ and Σx.



Diameter of coin **x** (cm)	2.2	2.5	2.6	2.65	2.9

n [1]

$\Sigma x/n$ [2]

$3\overline{x}+1$ [3]

Σx [4]

$\overline{x}$ [5]

7 A naturalist is investigating how many eggs the endangered kakapo bird lays in each brood cycle. The results are given in this frequency table.

Number of eggs	1	2	3
Frequency	7	p	2

If the mean number of eggs is 1.5, find the value of p.

[1]

Mean for Ungrouped Frequency Tables Using Σ

The number of children x in different families is recorded.

Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

Determine the mean number of children per family, $\overline{x}$.

Since the frequency tells us how many times each value occurs, the data when listed out in full would be as follows:

0 0 0 0 0 0 1 1 1 1
1 1 1 1 2 2 2 2 2 2
2 2 2 2 2 2 2 2 3 3
3 3 3 3 3 3 3 4 4 5

The value of 2 appears 14 times. Therefore, the total of these 2's is

$$fx = 14 \times 2 = 28$$

Σfx therefore represents the total of all these products, and thus the sum of x.

The number of values is the **total frequency**, i.e. $\mathbf{n = \Sigma f}$ Therefore:

For a grouped frequency table:

$$\overline{x} = \frac{\Sigma fx}{\Sigma f}$$

Example

The age of dogs at an animal shelter is recorded:

Age (x)	1	2	3	4	5
Frequency (f)	3	1	0	6	7

Determine the mean age $\overline{x}$.

$$\Sigma fx = (1 \times 3) + (2 \times 1) + (3 \times 0) + (4 \times 6) + (5 \times 7)$$

$$= 64$$

$$\Sigma f = 17$$

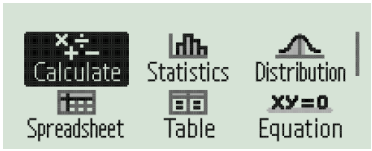
$$\overline{x} = \frac{\Sigma fx}{\Sigma f} = \frac{64}{17} = \mathbf{3.8 \text{ years}}$$

Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

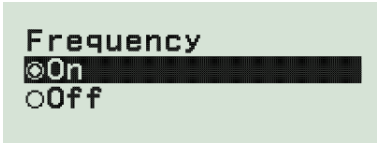
These are instructions for the **Casio fx-570/991CW**



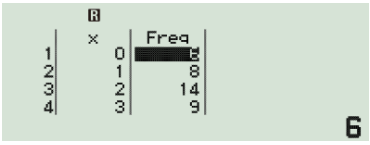
1. Use the arrows and OK to select Statistics. Choose 1-Variable.



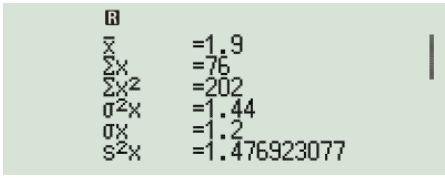
2. Press **Tools** than choose **Frequency**. Choose **On**.



3. Enter the values in x and their frequencies in Freq using EXE. Press OK for results.



4. Select **1-Var Results** and read off the values of $\overline{x}$. Note that **Σx represents Σfx** (since the x in Σx means the original data with duplicate values considered)

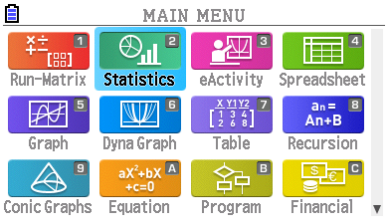


Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

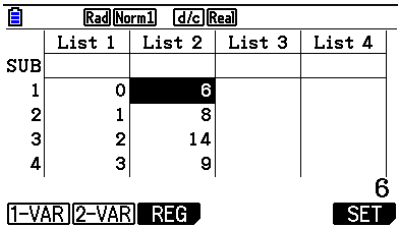
These are instructions for the **Casio fx-CG50**



1. Choose 2 for Statistics.



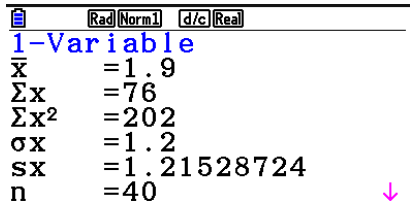
2. Enter the values of x and frequencies in the two columns, pressing EXE after each entry.



3. Press F2 (CALC), then F6 (SET). Set 1Var Freq to List 2 and press EXIT.



Press F1 for 1-Var.

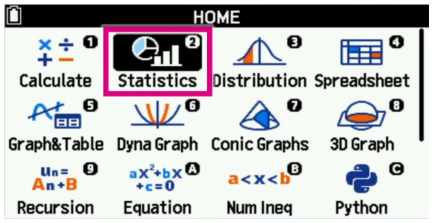


Num children (x)	0	1	2	3	4	5
Frequency (f)	6	8	14	9	2	1

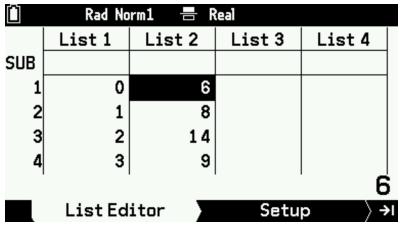
These are instructions for the **Casio fx-CG100**



1. Choose 2 for Statistics.



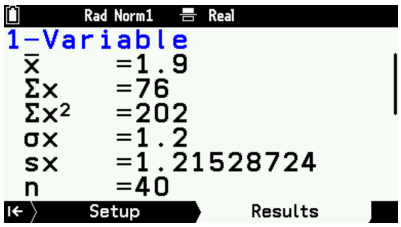
2. Enter the values of x and frequencies in the two columns, pressing EXE after each entry.



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and List 2 for Freq. Select Execute.



4. Read off the statistics



The age of members of a youth choir is recorded:

Age (x)	10	11	12	13	14	15
Frequency (f)	1	3	4	11	7	4

Determine the mean age $\overline{x}$.

1

Example: The weights of 40 randomly caught cats are summarised in the table below.

Weight (kg)	x	Number of cats, f
5 – 6	5.5	8
6 – 8	7	12
8 – 10	9	15
10 – 12	11	5

This time the original weights are not known, only the group each appears in.

Our 'best guess' is that each value will be, on average, the midpoint of the range it belongs to (e.g. we assume the 5-6kg cats were each 5.5kg). **We therefore replace each value with this midpoint x.**
In this way, Σfx represents the **estimated** total of the cat's weights, rather than the true (unknown) total.

Given that x denotes the midpoint of each group in the table and $\mathbf{\Sigma fx = 318}$
calculate an estimate for the mean.

If calculating from scratch:

$$\Sigma fx = (8 \times 5.5) + (12 \times 7) + (15 \times 9) + (5 \times 11) = 318$$

Then using the normal formula:

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{318}{40} = 7.95 \text{ kg}$$

Actual Exam Question:
The times, in seconds, spent in a queue at a supermarket by 85 randomly selected customers, are summarised in the table below.

Time (seconds)	Number of customers, f
0 – 30	2
30 – 60	10
60 – 70	17
70 – 80	25
80 – 100	25
100 – 150	6

1

Given that x denotes the midpoint of each group in the table and
$\Sigma fx = 6460 \quad \Sigma fx^2 = 529400$
calculate an estimate for the mean.

(1 mark)

NOTE: Σfx^2 is only needed if calculating standard deviation, which is covered later.

The heights of plants, in centimetres, in a garden centre are given in the table below.

Height (cm)	Number of plants, f
0 – 10	7
10 – 20	3
20 – 50	8
50 – 100	12

Given that h denotes the midpoint of each group in the table, calculate an estimate for the mean height $\overline{h}$.

1

2 The noise levels at 30 locations near an outdoor concert venue were measured to the nearest decibel. The data collected is shown in the grouped frequency table.

Noise (decibels)	65–69	70–74	75–79	80–84	85–89	90–94	95–99
Frequency	1	4	6	6	8	4	1

a Calculate an estimate of the mean noise level.

b Explain why your answer to part **a** is an estimate.

a

b

3 The table shows the daily mean temperature at Heathrow in October 1987 from the large data set.

Temp (°C)	$6 \leqslant t < 8$	$8 \leqslant t < 10$	$10 \leqslant t < 12$	$12 \leqslant t < 14$	$14 \leqslant t < 16$	$16 \leqslant t < 18$
Frequency	3	7	9	7	3	2

a Write down the modal class.

b Calculate an estimate for the mean daily mean temperature.

a

b

4 Two DIY shops (A and B) recorded the ages of their workers.

Age of worker	16–25	26–35	36–45	46–55	56–65	66–75
Frequency A	5	16	14	22	26	14
Frequency B	4	12	10	28	25	13

By comparing estimated means for each shop, determine which shop is better at employing older workers.

1

The table shows the heights of various cats.

Height **h** (cm)	Frequency
$0 \leq h \leq 10$	8
$10 \leq h \leq 20$	60

Estimate the number of cats with a height:

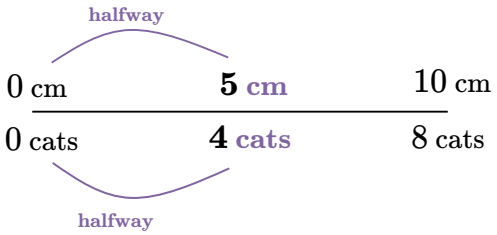
a. Below 5 cm
b. Above 15 cm
c. Below 12.5 cm

a

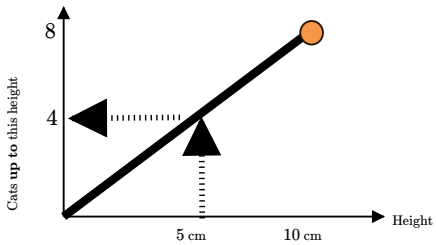
b

c

We are assuming the 8 cats are **equally distributed** between 0 and 10 cm, so that there are 4 between 0 – 5 cm and 4 between 5 – 10 cm.



This is known as **linear interpolation**, because if we were plotting this as a cumulative frequency graph (i.e. the running total of cats up to a specific height), **the graph forms a straight line**.



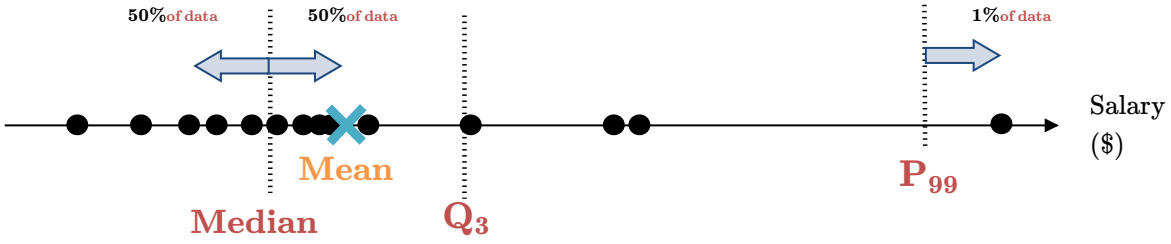
The grouped frequency table shows the times taken for students to complete a puzzle.

Time **t** (secs)	Frequency
$0 \leq t < 20$	12
$20 \leq h < 40$	8
$40 \leq h < 50$	20

Estimate the number of students with a time above 25 seconds.

1

MEAN: *Average* that considers all values, but is skewed by extreme values.



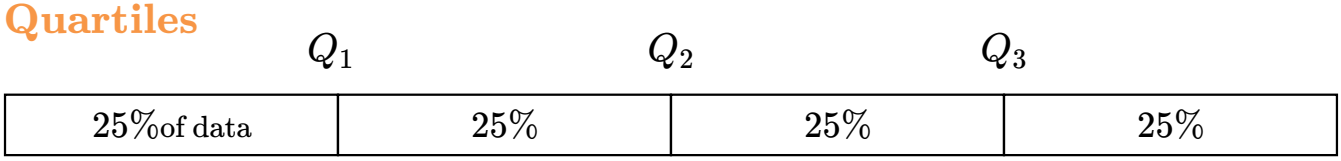
MEDIAN: a measure of position because it allows us to get the value a certain position, in this case 50%,along the data.

QUARTILE: gives us the value 25%, 50%, and 75%along the data.
$Q_1 = 25\%$ along data (lower quartile)
$Q_3 = 75\%$ along data (upper quartile)

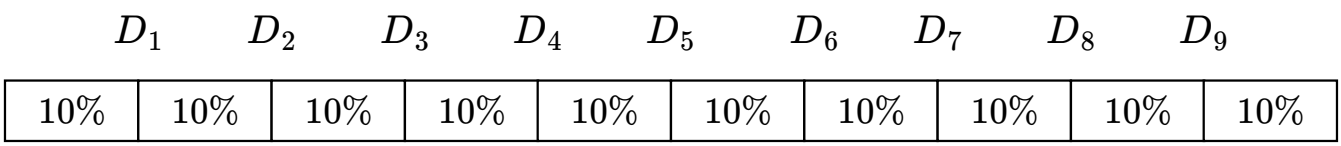
PERCENTILE: % value along the data. For example, the 99th percentile (P_{99}) is the value 99%along the data. When 'the 1%'is used in the media, it's referring to people in the 'top percentile' for salary, i.e. above P_{99}.

Quantiles represent general positions across the data when ordered.

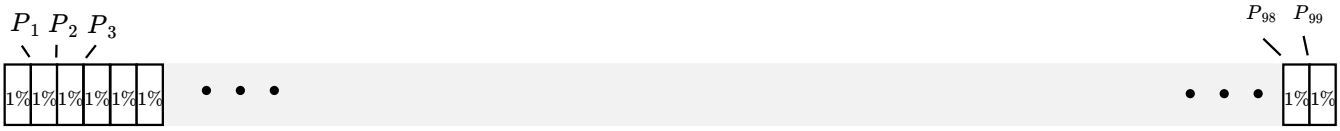
Quartiles



Deciles



Percentiles



4,9,10,15	4	1	2
		3	4
1,2,3,5,6,6,9,9,10,11,12	10	5	6
		7	8

Can you think of a rule to find the position of the median given the number of values, n?

The median is the $\frac{1}{2}(n+1)$ th item.
If the position is a decimal, use interpolation.

e.g. the '2.5th' item is halfway between the 2nd and 3rd.
The '4.25th' item is a quarter of the way between the 4th and 5th.

Which Item To Use For Grouped Data? 27

IQ of L6Ms2 ($\mathbf{q}$)	Frequency ($\mathbf{f}$)
$80 \leq q < 90$	7
$90 \leq q < 100$	5
$100 \leq q < 120$	3
$120 \leq q < 200$	2

If the data is grouped, what item do we use for the median?

median

To estimate the median of grouped data, calculate $\frac{n}{2}$, then use linear interpolation.

Notice we use $\frac{n}{2}$ rather than $\frac{n+1}{2}$ if grouped data.

Quickfire Questions 28

What position do we use for the median?

1. Lengths: 3cm, 5cm, 6cm, ...
$n = 11$

Median position: **6th** [2]

2. Lengths: 4m, 8m, 12.4m, ...
$n = 24$

Median position: **12th** [4]

3. Age	Freq
$10 \leq a < 20$	12
$20 \leq a < 30$	5

Median position: [8]

4. Score	Freq
$150 \leq s < 200$	3
$200 \leq s < 400$	7

Median position: [3]

5. Ages: 5, 7, 7, 8, 9, 10, ...
$n = 60$

Median position: **30th** [5]

6. Score	Freq
$150 \leq s < 200$	15
$200 \leq s < 400$	6

Median position: [9]

7. Weights: 1.2kg, 3.3kg, ...
$n = 35$

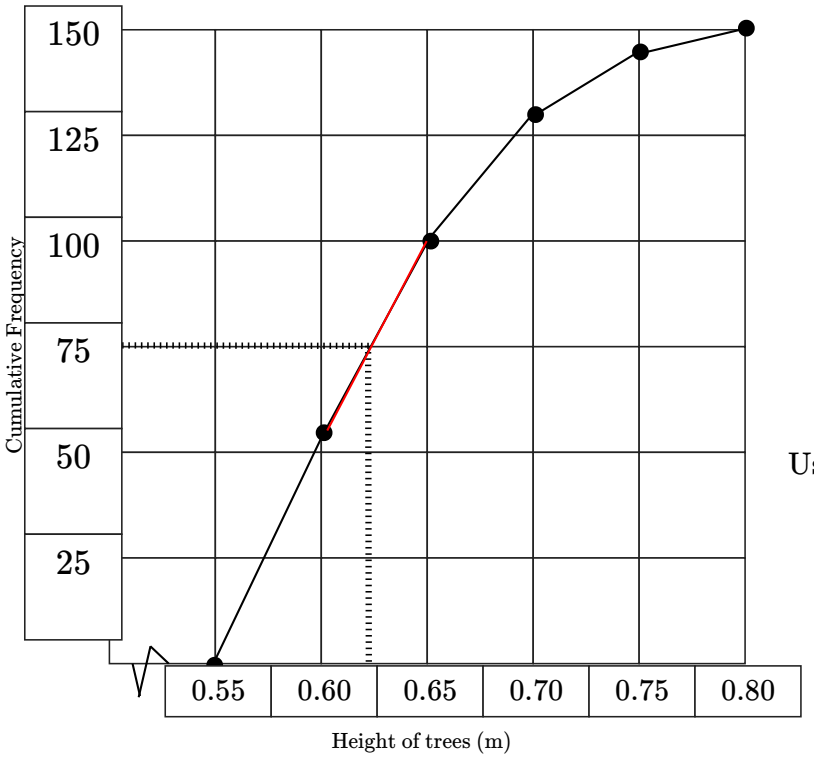
Median position: **18th** [1]

8. Volume (ml)	Freq
$0 \leq v < 100$	5
$100 \leq v < 200$	6
$200 \leq v < 300$	2

Median position: [6]

9. Weights: 4.4kg, 7.6kg, 7.7kg...
$n = 18$

Median position: **9th**/ [7]

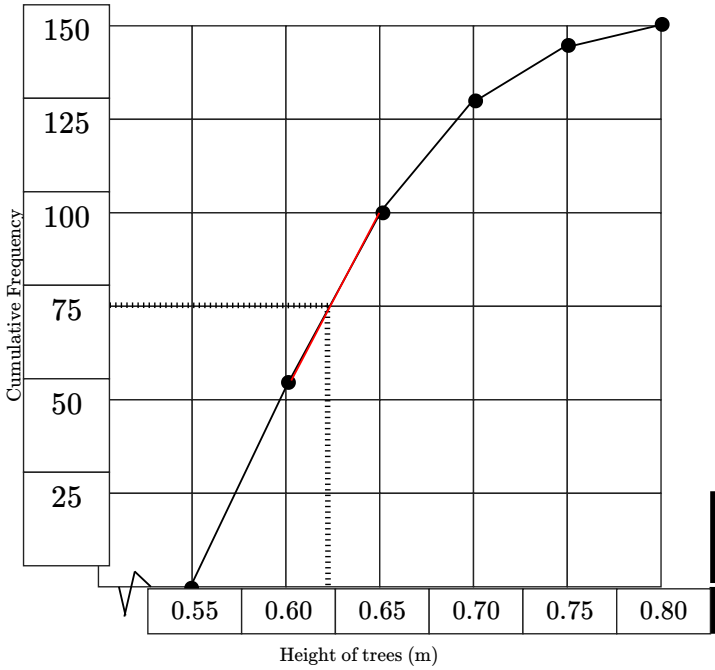


Height of tree (m)	Freq	C.F
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

Using a cumulative frequency graph, we know we can estimate the median by drawing a **suitable line**.

How could we read off this value exactly using a suitable calculation?

We could find the fraction of the way along the line segment using the frequencies, then go this same fraction along the class interval.



Height of tree (m)	Freq	C.F
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

Use linear interpolation to estimate the median.

Step 1: Identify the **interval** in which the median item, here the $\frac{150}{2} = 75^{\text{th}}$ value, lies.

Step 2: Write the relevant data for the calculation

Frequency up until this interval	Item number we're interested in	Frequency by end of this interval
Height at start of interval	Quartile	Height by end of interval

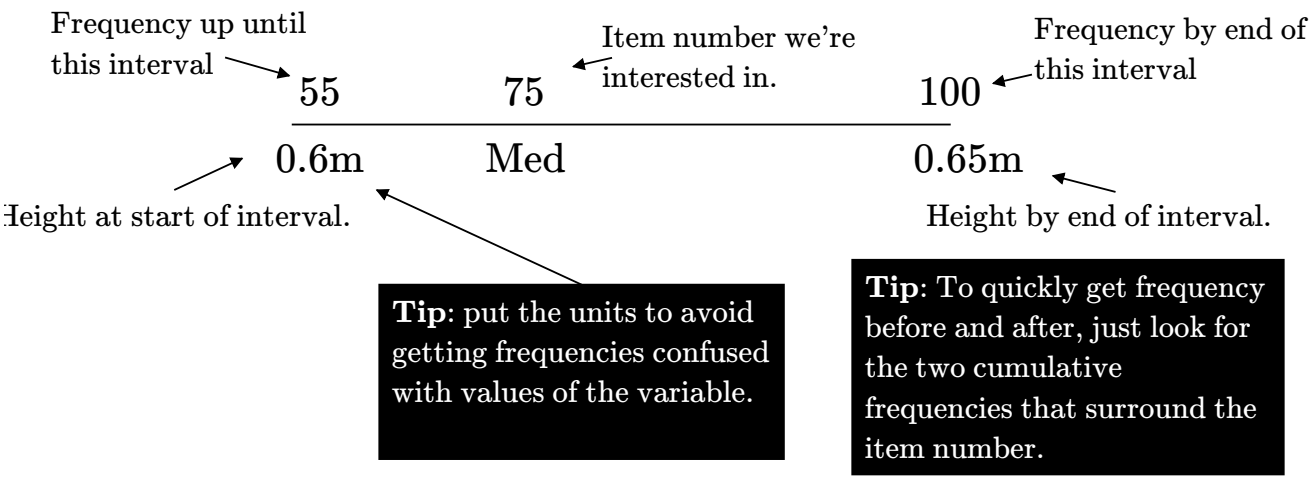
Height of tree (m)	Freq	C.F.
$0.55 \leq h < 0.6$	55	55
$0.6 \leq h < 0.65$	45	100
$0.65 \leq h < 0.7$	30	130
$0.7 \leq h < 0.75$	15	145
$0.75 \leq h < 0.8$	5	150

What fraction of the way across the interval are we?

[1]

Hence:

Median = [2]



Tip: put the units to avoid getting frequencies confused with values of the variable.

Tip: To quickly get frequency before and after, just look for the two cumulative frequencies that surround the item number.

1. Weight of cat (kg)	Freq	C.F.
$1.5 \leq w < 3$	10	10
$3 \leq w < 4$	8	18
$4 \leq w < 6$	14	32

Median class interval: [1]

[5] [6] [7]

[10] Med [11]

Fraction along interval: [12]

Median: [14]

2. Time (s)	Freq	C.F.
$8 \leq t < 10$	4	4
$10 \leq t < 12$	3	7
$12 \leq t < 14$	13	20

[2] [3] [4]

[8] Med [9]

Median:

[13]

What's different about the interval here? 33

Weight of cat to nearest kg	Frequency
$10 - 12$	7
$13 - 15$	2
$16 - 18$	9
$19 - 20$	4

There are **GAPS** between intervals!
What interval does this **actually** represent?

$$10 - 12$$

Lower class boundary — upper class boundary

Class Width

Identify the class width 34

1. Distance **d** travelled (in m)	...
$0 \leq d < 150$	
$150 \leq d < 200$	
$\mathbf{200 \leq d < 210}$	

Lower class boundary = [1]

Class width = [3]

3. Time **t** taken (in seconds)	...
$0 - 3$	
$\mathbf{4 - 6}$	
$7 - 11$	

Lower class boundary = [2]

Class width = [4]

2. Weight **w** in kg	...
$10 - 20$	
$21 - 30$	
$\mathbf{31 - 40}$	

Lower class boundary = [5]

Class width = [7]

4. Speed **s** (in mph)	...
$10 \leq s < 20$	
$20 \leq s < 29$	
$\mathbf{29 \leq s < 31}$	

Lower class boundary = [6]

Class width = [8]

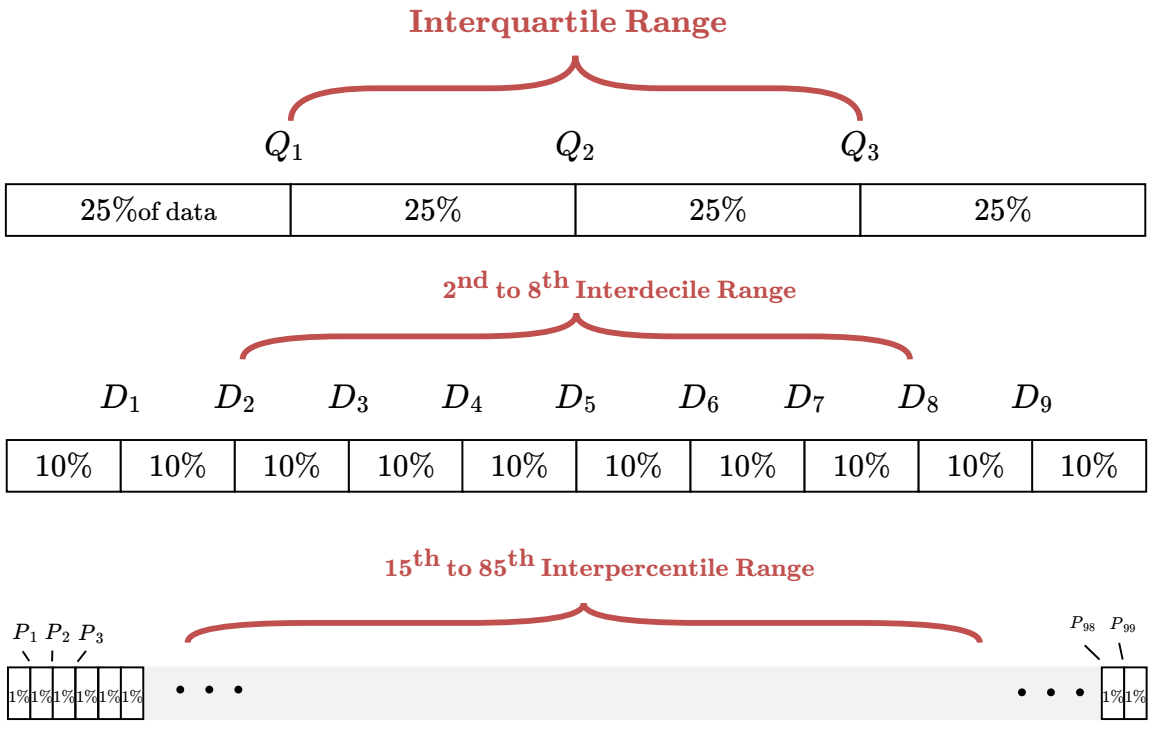
Example:

Summarised below are the distances, to the nearest mile, travelled to work by a random sample of 120 commuters.

Use linear interpolation to estimate the distribution's median

Distance (to the nearest mile)	Number of commuters	
0 – 9	10	10
10 – 19	19	29
20 – 29	43	72
30 – 39	25	97
40 – 49	8	105
50 – 59	6	111
60 – 69	5	116
70 – 79	3	119
80 – 89	1	120

$$\frac{[1] \qquad [2] \qquad [3]}{[5] \qquad \text{Med} \qquad [6]} \qquad Median = [4]$$



Just as we can calculate the interquartile range to mean the range of the middle 50% of data, we can also use deciles and percentiles to find the range of a more general middle percentage of the data.

Use linear interpolation to estimate the 3rd to 7th interdecile range of the following data:

Mass (x) to the nearest gram	Freq ($\mathbf{f}$)
10 – 12	5
13 – 14	15
15 – 18	7
19 – 22	3

1

3 A hotel is worried about the reliability of its lift. It keeps a weekly record of the number of times it breaks down over a period of 26 weeks. The data collected is summarised in the table opposite.

Use interpolation to estimate the median number of breakdowns.

Number of breakdowns	Frequency
0–1	18
2–3	7
4–5	1

(2 marks)

1

The number of patients attending a hospital trauma clinic each day was recorded over several months, giving the data in the table below.

Number of patients	10 - 19	20 - 29	30 - 34	35 - 39	40 - 44	45 - 49	50 - 69
Frequency	2	18	24	30	27	14	5

Use linear interpolation to estimate the median of these data.

1

The ages of 300 houses in a village are recorded given the following table of results.

Age a (years)	Number of houses
$0 \leq a < 20$	36
$20 \leq a < 40$	92
$40 \leq a < 60$	74
$60 \leq a < 100$	39
$100 \leq a < 200$	14
$200 \leq a < 300$	27
$300 \leq a < 500$	18

Use linear interpolation to estimate the median.

2

A cyber-café recorded how long each user stayed during one day giving the following results.

Length of stay (minutes)	Number of houses
$0 \leq l < 30$	15
$30 \leq l < 60$	31
$60 \leq l < 90$	32
$90 \leq l < 120$	23
$120 \leq l < 240$	17
$240 \leq l < 360$	2

Use linear interpolation to estimate the median of these data.

3

4. The following table summarises the times, t minutes to the nearest minute, recorded for a group of students to complete an exam.

Time (minutes) t	11 – 20	21 – 25	26 – 30	31 – 35	36 – 45	46 – 60
Number of students f	62	88	16	13	11	10

[You may use $\sum ft^2 = 134281.25$]

a) Estimate the mean of this data

b) Use linear Interpolation to estimate the value of the median

a

b

You need to be able to find the **quartiles** of both listed data and of grouped data. The rule is exactly the same as for the median.

Listed data

4,8,10,15	4	1	2
		3	4
1,2,3,5,6,8,9,9,10,11,12	10	5	6
		7	8

Can you think of a rule to find the position of the LQ/UQ given n?

9

10

The LQ, median and UQ give you 25%, 50% and 75% along the data respectively.

$n = 43$

Item to use for 57th percentile?



1

You will always find these for grouped data in an exam, **so never round this position**.

Notation:

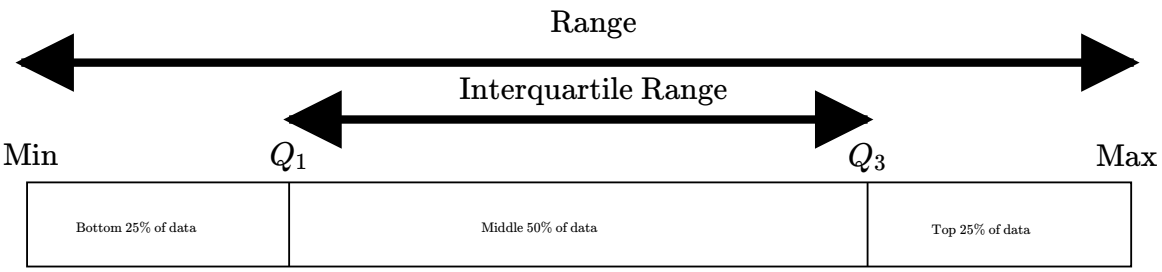
Lower Quartile: 2

Median: 3

Upper Quartile: 4

57th Percentile: 5

The interquartile range and interpercentile range are examples of **measures of spread.**

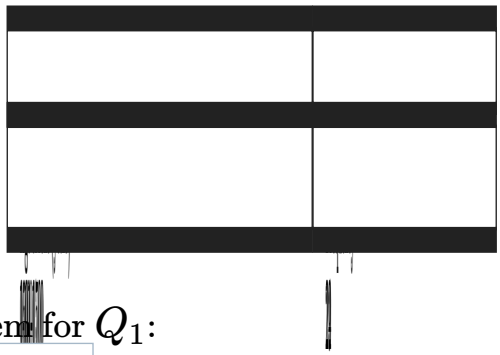


Interquartile Range = Upper Quartile − Lower Quartile

Why might we favour the interquartile range over the range?

1

These are the same as the 'Test Your Understanding' questions on your supplementary sheet from before.



Item for Q_1:

2

3

6

7

Item to use for P_{10}:

1

4

5

10th to 90th interpercentile range:

8

5 A roadside assistance firm kept a record over a week of the amount of time, in minutes, people were kept waiting for assistance. The times are shown below.

Time waiting, t (minutes)	$20 \leqslant t < 30$	$30 \leqslant t < 40$	$40 \leqslant t < 50$	$50 \leqslant t < 60$	$60 \leqslant t < 70$
Frequency	6	10	18	13	2

a Find an estimate for the mean wait time. **(1 mark)**

b Calculate the 65th percentile. **(2 marks)**

The firm writes the following statement for an advertisement:

> Only 10% of our customers have to wait longer than 56 minutes.

c By calculating a suitable percentile, comment on the validity of this claim. **(3 marks)**

a

b

c

6 The table shows the recorded wingspans, in metres, of 100 endangered Californian condors.

Wingspan, w (m)	$1.0 \leqslant w < 1.5$	$1.5 \leqslant w < 2.0$	$2.0 \leqslant w < 2.5$	$2.5 \leqslant w < 3.0$	$3.0 \leqslant w$
Frequency	4	20	37	28	11

a Estimate the 80th percentile and interpret the value. **(3 marks)**

b State why it is not possible to estimate the 90th percentile. **(1 marks)**

a

b

2 The table shows the monthly income for workers in a factory.

Monthly income, x (£)	$900 \leqslant x < 1000$	$1000 \leqslant x < 1100$	$1100 \leqslant x < 1200$	$1200 \leqslant x < 1300$
Frequency	3	24	28	15

a Calculate the 34% to 66% interpercentile range. **(3 marks)**

b Estimate the number of data values that fall within this range. **(2 marks)**

a

b

3 A train travelled from Lancaster to Preston. The times, to the nearest minute, it took for the journey were recorded over a certain period. The times are shown in the table.

Time for journey (minutes)	15–16	17–18	19–20	21–22
Frequency	5	10	35	10

a Calculate the 5% to 95% interpercentile range. **(3 marks)**

b Estimate the number of data values that fall within this range. **(1 mark)**

a

b

4 From the large data set, the daily mean temperature (°C) for Leeming during the first 10 days of June 1987 is given below:

14.3 12.7 12.4 10.9 9.4 13.2 12.1 10.3 10.3 10.6

a Calculate the median and interquartile range. **(2 marks)**

The median daily mean temperature for Leeming during the first 10 days of May 1987 was 9.9 °C and the interquartile range was 3.9 °C.

b Compare the data for May with the data for June. **(2 marks)**

The 10% to 90% interpercentile range for the daily mean temperature for Leeming during July 1987 was 5.4 °C.

c Estimate the number of days in July 1987 on which the daily mean temperature fell within this range. **(1 mark)**

a

b

c

Q1) May 2013 Q4 (continued)

The following table summarises the times, t minutes to the nearest minute, recorded for a group of students to complete an exam.

Time (minutes) t	11 – 20	21 – 25	26 – 30	31 – 35	36 – 45	46 – 60
Number of students f	62	88	16	13	11	10

(*c*) Show that the estimated value of the lower quartile is 18.6 to 3 significant figures. **(1)**

(*d*) Estimate the interquartile range of this distribution. **(2)**

1

Q2) June 2005 Q2

The following table summarises the distances, to the nearest km, that 134 examiners travelled to attend a meeting in London.

Distance (km)	Number of examiners
41–45	4
46–50	19
51–60	53
61–70	37
71–90	15
91–150	6

(*c*) Use interpolation to estimate the median Q_2, the lower quartile Q_1, and the upper quartile Q_3 of these data.

2

Q3)

The ages of 300 houses in a village are recorded given the following table of results.

Age a (years)	Number of houses
$0 \leq a < 20$	36
$20 \leq a < 40$	92
$40 \leq a < 60$	74
$60 \leq a < 100$	39
$100 \leq a < 200$	14
$200 \leq a < 300$	27
$300 \leq a < 500$	18

Use linear interpolation to estimate the lower quartile, upper quartile and hence the interquartile range.

1

Q4)

A cyber-café recorded how long each user stayed during one day giving the following results.

Length of stay (minutes)	Number of houses
$0 \leq l < 30$	15
$30 \leq l < 60$	31
$60 \leq l < 90$	32
$90 \leq l < 120$	23
$120 \leq l < 240$	17
$240 \leq l < 360$	2

Use linear interpolation to estimate:

a) The lower quartile.
b) The upper quartile.
c) The 90th percentile.

3

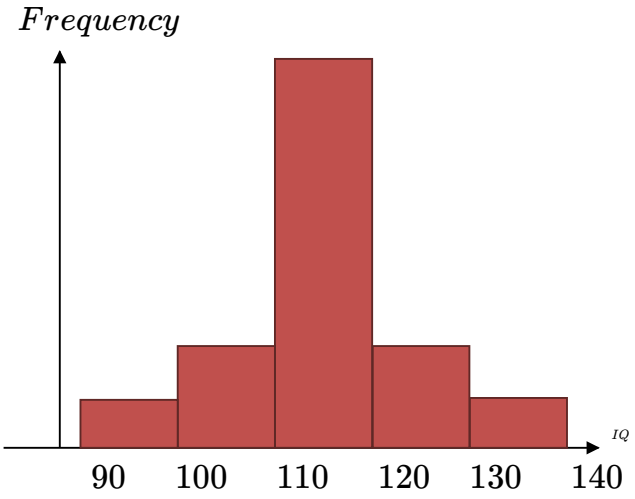
Q5)

Distance (to the nearest mile)	Number of commuters
0 – 9	10
10 – 19	19
20 – 29	43
30 – 39	25
40 – 49	8
50 – 59	6
60 – 69	5
70 – 79	3
80 – 89	1

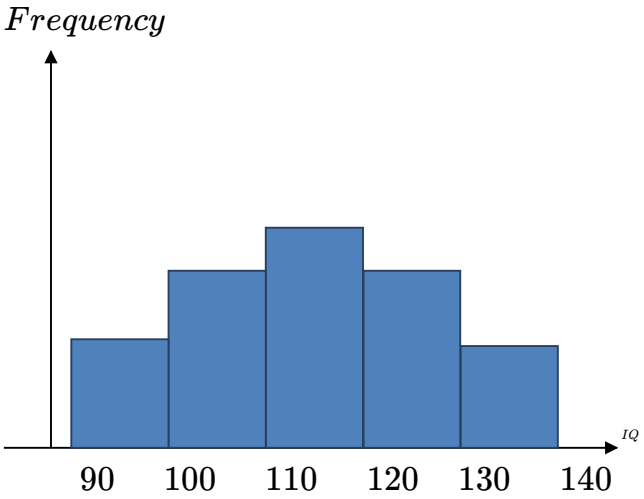
Find the interquartile range for the distance travelled by commuters.

2

Distribution of IQs in Class A



Distribution of IQs in Class B



Here are the distribution of IQs in two classes of the same size. What's the same, and what's different?

1

Variance is a measure of spread that takes all values into account.
Variance, by definition, is the **average squared distance from the mean**.

$$\sigma^2 = \frac{\sum(x - \overline{x})^2}{n}$$

Notation: While Σ is 'uppercase sigma' and means 'sum of', σ is 'lowercase sigma' (we'll see why we have the squared in a sec)

Distance from mean. . .

Squared distance from mean. . .

Average squared distance from mean. . .

In practice, it is possible to simplify the formula to the following*:

Variance mean of the squares minus the square of the mean:

$$\sigma^2 = \square^{1} - \square^{2}$$

*** Proof: (not in syllabus)**

3

Standard Deviation

$$\sigma = \sqrt{Variance}$$

The standard deviation **can 'roughly' be thought of as the average distance from the mean.**

3, 11

Variance

$\sigma^2 =$ [1]

Standard Deviation

$\sigma =$ [4]

2cm 3cm 3cm 5cm 7cm

Variance

$\sigma^2 =$ [2]

Standard Deviation

$\sigma =$ [3]

Find the variance and standard deviation of the following sets of data.

2 4 6

Variance = [1]

Standard Deviation = [2]

1 2 3 4 5

Variance = [3]

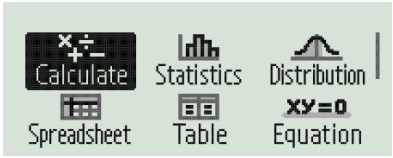
Standard Deviation = [4]

Time (secs)	30	34	35	39

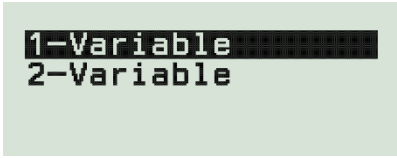
These are instructions for the **Casio fx-570/991CW**



1. Use the arrows and OK to select Statistics.



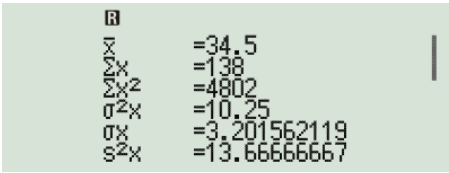
2. Choose 1-Variable.



3. Enter each value with EXE. After the last entry press OK to open the menu.



4. Choose '1 Variable Results'. Read off σx (σ) and $\sigma^2 x$ (the variance σ^2)

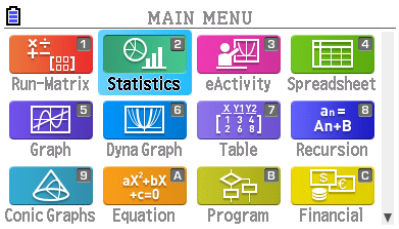


Time (secs)	30	34	35	39

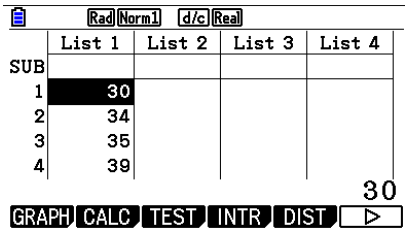
These are instructions for the **Casio fx-CG50**



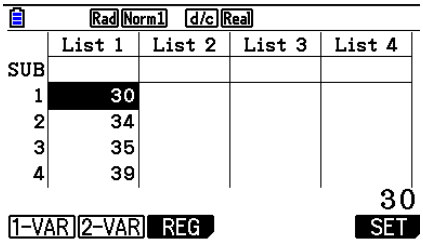
1. Choose 2 for Statistics.



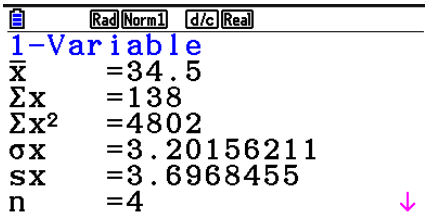
2. Enter your values, pressing EXE after each and using the arrow keys to move around.



3. In CALC (F2), open SET (F6). Set 1Var Freq to 1, press EXIT, then 1-VAR (F1).



4. Read off σ.

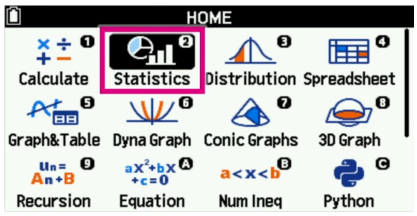


Time (secs)	30	34	35	39

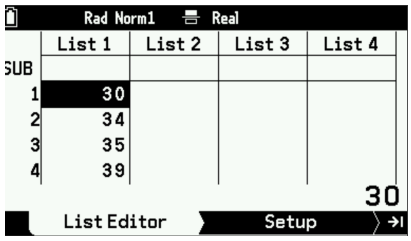
These are instructions for the **Casio fx-CG100**



1. Choose 2 for Statistics.



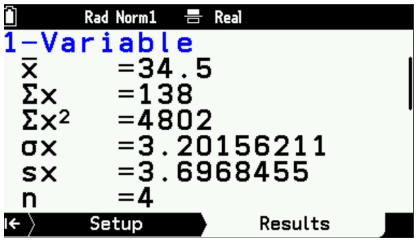
2. Enter your values, pressing EXE after each and using the arrow keys to move around.



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and 1 for Freq. Select Execute.



4. Read off σ.



Use a calculator to determine the standard deviation of the following data set of long jump distances.

6.1 m, 8.3 m, 8.3 m, 8.4 m, 8.5 m, 8.7 m

1

It is unlikely in an exam you would be asked to compute the standard deviation from scratch. Most likely you would calculate σ directly without formula or be given key summations to use in the formula.

EXAMPLE

Before going on holiday to Seapron, Tania records the weekly rainfall (x mm) at Seapron for 8 weeks during the summer. Her results are summarised as

$$\Sigma x = 86.8 \qquad \Sigma x^2 = 985.88$$

$$\sigma^2 = \frac{\Sigma x^2}{n} - \overline{x}^2$$

Find the standard deviation, σ_x, for these data.

(3 marks)

1. Calculate $\overline{x}$ using $\frac{\Sigma x}{n}$:

Mean

2. Use the formula with provided Σx^2:

Variance

3. Calculate σ_x :

σ

[Edexcel AS SAM P2 Q1c]

Sara is investigating the variation in daily maximum gust, t kn, for Camborne in June and July 1987. She used the large data set to select a sample of size 20 from the June and July data for 1987.

Sara selected the first value using a random number from 1 to 4 and then selected every third value after that.

The data Sara collected are summarised as follows

$$\mathrm{n} = 20 \qquad \Sigma \mathrm{t} = 374 \qquad \Sigma \mathrm{t}^2 = 7600$$

Calculate the standard deviation.

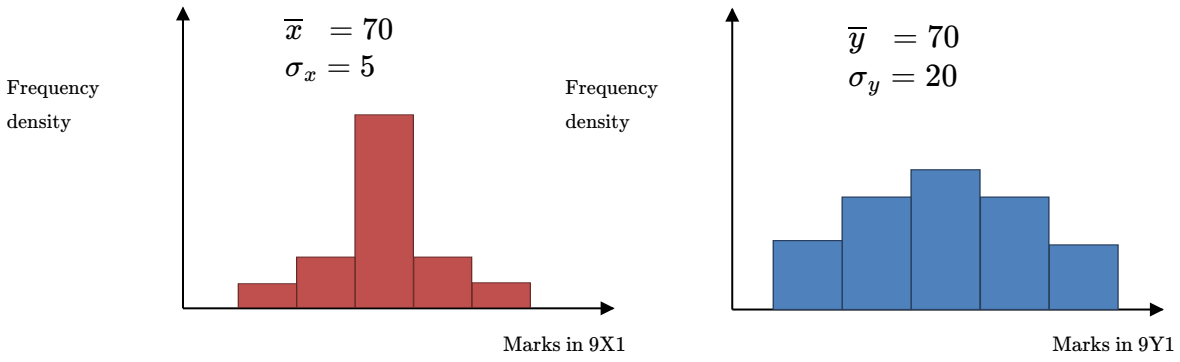
(2 marks)

Mark scheme

You may want to use the following mnemonic:

Variance: "The mean of the squares minus the square of the mean ('msmsm')"

$$\text{Variance} = \square^{1} \quad \square^{2}$$



Compare the marks of the two classes.

The classes had the same mean mark, but 9X1 had a lower standard deviation, suggesting the marks were **more consistent**.

Note: When σ is lower, it means there's less variation, which means **more consistency**.

Previously when comparing two data sets, we might compare:

1. A measure of **central tendency** (usually the mean)
2. A measure of **spread**

For the latter, you may have previously used the range or interquartile range. **Now you can use the standard deviation**!

Exam Question

The number of hours of sunshine each day, y, for the month of July at Heathrow are summarised in the table below.

Hours	$0 \leqslant y < 5$	$5 \leqslant y < 8$	$8 \leqslant y < 11$	$11 \leqslant y < 12$	$12 \leqslant y < 14$
Frequency	12	6	8	3	2

It can be shown that the mean of this data is 6.63 and that the standard deviation is 3.69.

The mean and standard deviation for the number of hours of daily sunshine for the same month in Hurn are 5.98 hours and 4.12 hours respectively.

Thomas believes that the further south you are the more consistent should be the number of hours of daily sunshine.

State whether or not the standard deviations for the two locations support Thomas' belief.

Mark Scheme

If we consider the standard deviation as, approximately, the average distance of each value from the mean, we can reflect on how additional values will impact the standard deviation.

The populations of 10 countries are recorded. The mean population is 9 million, and the standard deviation is 1 million.

I add Switzerland to my data set, which has a population of 9 million.

State the effect on:

a. the range of the data

b. the standard deviation of the data

a

b

Data is…	Mean	Variance
Ungrouped	$\bar{x} = \frac{\Sigma x}{n}$	$\sigma^2 = \frac{\Sigma x^2}{n} - \bar{x}^2$
Grouped	1	2

EXAM QUESTION

The following table summarises the times, t, minutes to the nearest minute, recorded for a group of students to complete an exam.

Time (minutes) t	11 – 20	21 – 25	26 – 30	31 – 35	36 – 45	46 – 60
Number of students f	62	88	16	13	11	10

[You may use $\Sigma ft^2 = 134281.25$*]*

Estimate the mean and standard deviation of these data. **(5)**

1. Use each class midpoint as its representative value. Find the six midpoints.

2. Estimate the mean.

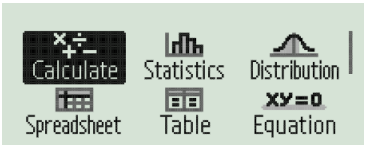
3. Estimate the standard deviation using $\sigma^2 = \Sigma ft^2 / \Sigma f - \bar{t}^2$.

Time (minutes) t	11 – 20	21 – 25	26 – 30	31 – 35	36 – 45	46 – 60
Midpoints	15.5	23	28	33	40.5	53
Number of students f	62	88	16	13	11	10

These are instructions for the **Casio fx-570/991CW**



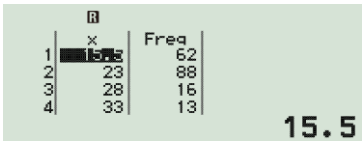
1. Use the arrows and OK to select Statistics.



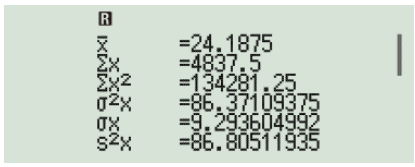
2. Choose 1-Variable. Open Tools, set Frequency to On, then press Back.



3. Enter the values in x and their frequencies in Freq using EXE. Press OK for results.



4. Choose '1 Variable Results'. Read off σx (σ) and $\sigma^2 x$ (the variance σ^2)



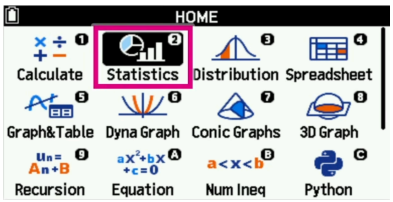
Important note: Σx^2 here means $\Sigma f x^2$, consistent with the value given in the question on the previous slide. The frequencies are implicitly 'baked in'.

Time (minutes) t	11 – 20	21 – 25	26 – 30	31 – 35	36 – 45	46 – 60
Midpoints	15.5	23	28	33	40.5	53
Number of students f	62	88	16	13	11	10

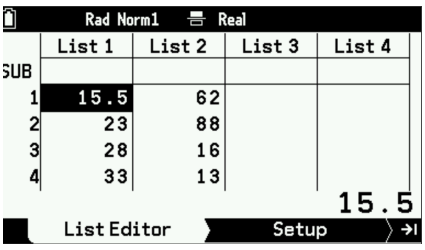
These are instructions for the **Casio fx-CG100**



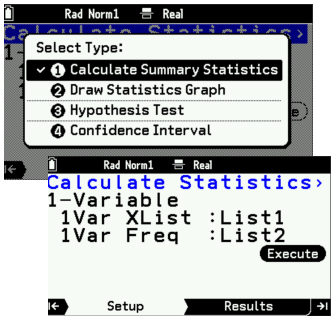
1. Choose 2 for Statistics.



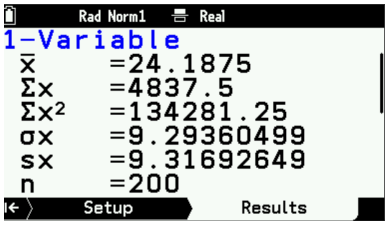
2. Enter the midpoints in List 1 and the Frequencies in List 2



3. Select Calculate Summary Statistics, then 1-Variable. Use List 1 for XList and List 2 for Freq. Select Execute.



4.Read off σx (σ). Square this value for the variance.



Exam Question

An agriculturalist is studying the yields, y kg, from tomato plants. The data from a random sample of 70 tomato plants is summarised below.

Yield (y kg)	Frequency (f)	Yield midpoint (x kg)
$0 \leq y < 5$	16	2.5
$5 \leq y < 10$	24	7.5
$10 \leq y < 15$	14	12.5
$15 \leq y < 25$	12	20
$25 \leq y < 35$	4	30

(You may use $\Sigma fx = 755$ and $\Sigma fx^2 = 12037.5$)

Estimate the mean and the standard deviation of the yields of the tomato plants.

Mark Scheme

Example: In classes 7A and 7B, consisting of 20 and 30 students respectively, the sums of their maths scores were $\Sigma a = 1200$ and $\Sigma b = 2100$, and $\Sigma a^2 = 75000$ and $\Sigma b^2 = 160000$.

Determine the standard deviation of maths scores in the two classes combined.

Let x be the scores across both classes.

The number of values, n, is just the total students across the two sets.

n

Similarly, the total score, Σx, across both classes is just the sum from each class:

Combined totals

Once σ^2 is found, calculate the standard deviation:

σ

Further Example

In classes 8A and 8B, consisting of 5 and 10 students respectively, the means of their maths scores were 60 and 70, and the standard deviations 6 and 7.

Determine the standard deviation across all 15 students.

1. Use $\overline{x} = \frac{\Sigma x}{n}$ formula to determine Σx

1

2. Similarly use σ^2 formula backwards to determine Σx^2

3

3. Repeat the above steps for class 8B

4

4. combine sums together.

σ

Test Your Understanding

Exam Question

Throughout the year Alice took nine written tests and five practical tests in Music.

The mean mark of her nine written tests was 69.

The mean mark of her five practical tests was 83.

The standard deviation of her nine written tests was 15.4.

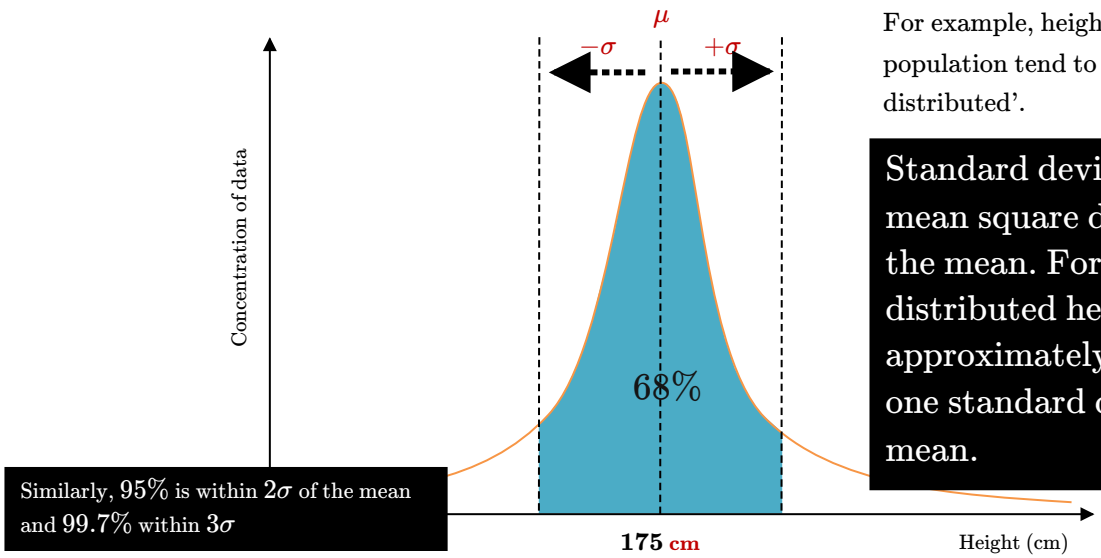
The standard deviation of her five practical tests was 7.8.

Calculate the standard deviation of all 14 tests.

(4 marks)

1

Standard deviation can be used as a measure of spread **for data with any 'shape'**. However, it becomes **particularly important when we consider normally distributed data**, i.e. has a symmetrical 'bell-curved' shape with data becoming less concentrated further away from the mean.



For example, heights in a population tend to be 'normally distributed'.

Standard deviation is the root mean square distance from the mean. For normally distributed heights, approximately 68% lie within one standard deviation of the mean.

Similarly, 95% is within 2σ of the mean and 99.7% within 3σ

You can find out about the normal distribution in skills 640-645.

2 Ten collie dogs are weighed (w kg). The summary data for the weights is:

$\Sigma w = 241$ $\Sigma w^2 = 5905$

Use this summary data to find the standard deviation of the collies' weights. **(2 marks)**

1

4 For a set of 10 numbers: $\Sigma x = 50$ $\Sigma x^2 = 310$

For a different set of 15 numbers: $\Sigma x = 86$ $\Sigma x^2 = 568$

Find the mean and the standard deviation of the combined set of 25 numbers.

2

5 Nahab asks the students in his year group how much pocket money they get per week. The results, rounded to the nearest pound, are shown in the table.

Number of £s	8	9	10	11	12
Frequency	14	8	28	15	20

a Use your calculator to work out the mean and standard deviation of the pocket money. Give units with your answer. **(3 marks)**

b How many students received an amount of pocket money more than one standard deviation above the mean? **(2 marks)**

a

b

6 In a student group, a record was kept of the number of days of absence each student had over one particular term. The results are shown in the table.

Number of days absent	0	1	2	3	4
Frequency	12	20	10	7	5

Use your calculator to work out the standard deviation of the number of days absent. **(2 marks)**

1

7 A certain type of machine contained a part that tended to wear out after different amounts of time. The time it took for 50 of the parts to wear out was recorded. The results are shown in the table.

Lifetime, h (hours)	$5 < h \leqslant 10$	$10 < h \leqslant 15$	$15 < h \leqslant 20$	$20 < h \leqslant 25$	$25 < h \leqslant 30$
Frequency	5	14	23	6	2

The manufacturer makes the following claim:

90% of the parts tested lasted longer than one standard deviation below the mean.

Comment on the accuracy of the manufacturer's claim, giving relevant numerical evidence.

2

8 The daily mean windspeed, x (kn) for Leeming is recorded in June 2015. The summary data is:

$\Sigma x = 243 \qquad \Sigma x^2 = 2317$

a Use your calculator to work out the mean and the standard deviation of the daily mean windspeed in June 2015. **(2 marks)**

The highest recorded windspeed was 17 kn and the lowest recorded windspeed was 4 kn.

b Estimate the number of days in which the windspeed was greater than one standard deviation above the mean. **(2 marks)**

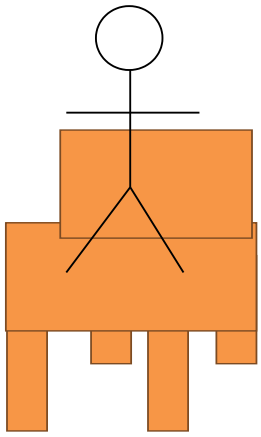
c State one assumption you have made in producing this estimate. **(1 mark)**

a

b

c

What do you reckon is the mean height of people in a room if they stand on a chair?



Is there an easy way to recalculate the mean based on the new heights? And the variance of the heights?

1

Suppose now after a bout of 'stretching you to your limits', you're now all 3 times your original height.

What do you think happens to the standard deviation of your heights?

1

What do you think happens to the variance of your heights?

2

Extension Question: Can you prove the latter using the formula for variance?

3

Suppose our original variable (e.g. heights in cm) was x.
Then y would represent the heights with 10cm added on to each value.

Coding	Effect on $\overline{x}$	Effect on σ
$y = x + 10$	1	2
$y = 3x$	3	4
$y = 2x - 5$	5	6

Cost x of diamond ring (£)

£1010 £1020 £1030 £1040 £1050

We 'code' our variable using the following:

$$y = \frac{x-1000}{10}$$

New values y:

1

Standard deviation of y (σ_y): 2

therefore...

Standard deviation of x (σ_x): 3

Old mean $\overline{x}$	Old σ_x	Coding	New mean $\overline{y}$	New σ_y
36	4	$y = x - 20$	1	2
3	4	$y = 2x$	72	16
35	4	$y = 3x - 20$	5	6
7	8	$y = \frac{x}{2}$	20	$\frac{3}{2}$
11	27	$y = \frac{x+10}{3}$	9	10
11	12	$y = \frac{x-100}{5}$	40	5

The following table summarises the times, t minutes to the nearest minute, recorded for a group of students to complete an exam.

Time (minutes) t	11 – 20	21 – 25	26 – 30	31 – 35	36 – 45	46 – 60
Number of students f	62	88	16	13	11	10

[You may use $\Sigma ft^2 = 134281.25$]

(*a*) Estimate the mean and standard deviation of these data. **(5)**
(*b*) Use linear interpolation to estimate the value of the median. **(2)**
(*c*) Show that the estimated value of the lower quartile is 18.6 to 3 significant figures. **(1)**
(*d*) Estimate the interquartile range of this distribution. **(2)**
(*e*) Give a reason why the mean and standard deviation are not the most appropriate summary statistics to use with these data. **(1)**

Suppose we've worked all these out already.

The person timing the exam made an error and each student actually took 5 minutes less than the times recorded above. The table below summarises the actual times.

Time (minutes) t	6 – 15	16 – 20	21 – 25	26 – 30	31 – 40	41 – 55
Number of students f	62	88	16	13	11	10

(*f*) Without further calculations, explain the effect this would have on each of the estimates found in parts (*a*), (*b*), (*c*) and (*d*). **(3)**

f)

1

3 The coded mean price of televisions in a shop was worked out. Using the coding $y = \frac{x - 65}{200}$ the mean price was 1.5. Find the true mean price of the televisions. **(2 marks)**

1

6 The weekly income, i, of 100 women workers was recorded.

The data was coded using $y = \frac{i - 90}{100}$ and the following summations were obtained:

$\Sigma y = 131, \ \Sigma y^2 = 176.84$

Estimate the standard deviation of the actual women workers' weekly income. **(2 marks)**

2

7 A meteorologist collected data on the annual rainfall, x mm, at six randomly selected places.

The data was coded using $s = 0.01x - 10$ and the following summations were obtained:

$\Sigma s = 16.1, \ \Sigma s^2 = 147.03$

Work out an estimate for the standard deviation of the actual annual rainfall. **(2 marks)**

1

9 From the large data set, the daily mean pressure, p hPa, in Hurn during June 2015 is recorded.

The data is coded using $c = \frac{p}{2} - 500$ and the following summary statistics are obtained:

$n = 30 \qquad \bar{c} = 10.15 \qquad S_{cc} = 296.4$

Find the mean and standard deviation of the daily mean pressure. **(4 marks)**

2

8 A teacher standardises the test marks of his class by adding 12 to each one and then reducing the mark by 20%.

If the standardised marks are represented by t and the original marks by m:

a write down a formula for the coding the teacher has used. **(1 mark)**

The following summary statistics are calculated for the standardised marks:

$n = 28 \qquad \bar{t} = 52.8 \qquad S_{tt} = 7.3$

b Calculate the mean and standard deviation of the original marks gained. **(3 marks)**

a

b

I have a list of 30 heights in the class. What item do I use for:

- Q_1? 8th 1
- Q_2? Bet 2
- Q_3? 23rd 3

For the following grouped frequency table, calculate:

Height **h** of bear (in metres)	Frequency
$0 \leq h < 0.5$	4
$0.5 \leq h < 1.2$	20
$1.2 \leq h < 1.5$	5
$1.5 \leq h < 2.5$	11

a) The estimate mean: 4

b) The estimate median: 5

c) The estimate variance:
(you're given $\Sigma fh^2 = 67.8125$) 6

What is the standard deviation of the following lengths: 1cm, 2cm, 3cm

1

The mean of a variable x is 11 and the variance 4.
The variable is coded using $y = \frac{x+10}{3}$. What is:

a) The mean of y? 2
b) The variance of y? 3

A variable x is coded using $y = 4x - 5$.
For this new variable y, the mean is 15 and the standard deviation 8.
What is:

c) The mean of the original data? 4
d) The standard deviation of the original data? 5